Project Documentation: Smartphone Market Analysis & KPI Dashboard

Project Name: Global Smartphone Market & Apple Performance Dashboard

Platform: Tableau Desktop / Tableau Public

Status: Completed

1. Project Overview

This project involves the end-to-end recreation of a comprehensive Tableau dashboard and data story analyzing the global smartphone market. The analysis spans granular device specifications, internal brand performance (specifically Apple), and macro-level industry revenue trends. The final deliverable is a multi-view Tableau Story equipped with custom calculated fields, dual-axis charts, and geographic mapping.

2. Data Sources

The project utilizes multiple distinct CSV datasets to drive the visualizations:

- `apple_products.csv.csv` (KPIs and Model Specs)
- `Flipkart_Smartphone.csv` (Battery Types and Brand Pricing)
- `Model share.csv` (Internal Apple Model Share)
- `Country-wise share.csv` (Top Selling Models globally)
- `Quarterly-share.csv` (2018 Market Share snapshot)
- `Annual Revenue.csv` (Historical Revenue 2006-2022)
- `Market Penetration(iphone).csv` (Geographic data)

3. Visualization Breakdown

The dashboard consists of 9 distinct worksheets, each built to address specific business questions:

1. **KPI Text Table:** A high-level view of Apple's core metrics, utilizing Measure Values to display Discount Percentage, MRP, Star Rating, and Sale Price.
2. **Model Specification Table:** A detailed text matrix mapping iPhone models to their specific hardware (Processor, Front/Rear Camera, Color) and correlating it with retail pricing.
3. **Battery Type Distribution (Horizontal Bar Chart):** Analyzes the correlation between battery technologies (Lithium, Lithium-Ion, Lithium-Polymer) and the average original retail price of smartphones.
4. **Brand Price Comparison (Treemap):** A hierarchical view categorizing major smartphone brands by RAM configurations and price point dimensions to visualize market positioning.

5. **Model-Wise Share of iPhone (Packed Bubble Chart):** Illustrates the internal market share distribution across the iPhone 13 and iPhone 14 product lines.
6. **Country-Wise Best Selling Smartphone (Dual-Axis Combo Chart):** Combines a bar chart and a synchronized line chart to track the sales share percentages and growth momentum of premium models.
7. **Quarterly Share (Donut Chart):** A customized dual-axis pie chart acting as a snapshot of the competitive multi-brand smartphone market share landscape in 2018.
8. **Annual Revenue Year-Wise (Line Chart):** A continuous time-series visualization tracking the upward trajectory of industry revenue from 2006 to 2022.
9. **Global Market Share (Filled Map):** A geographic visualization filtering on the OnePlus Nord CE 2 Lite 5G to highlight its specific 4.00% market penetration within India.

4. Technical Implementations & Workarounds

To achieve the exact visual requirements from the raw data, several advanced Tableau techniques were utilized:

- **Calculated Fields:** Created custom metrics such as `Sales difference` (MRP - Sale Price) and hardcoded percentage metrics for specialized map labeling.
- **Geographic Role Assignment:** Converted standard text fields into recognized geographic dimensions to force map generation without native latitude/longitude columns.
- **Dual-Axis Synchronization:** Applied synchronized dual axes for both the combo chart (Bar/Line) and the Donut chart (Pie/Circle overlay).
- **Data Blending Avoidance:** Kept worksheets strictly separated by their respective data sources to avoid cross-database join errors, compiling them only at the final Dashboard/Story level.

5. Final Deliverables

- 9 individual Tableau Worksheets.
- 1 compiled Tableau Dashboard.
- 1 Tableau Story presentation, complete with a professional 2-minute voiceover script guiding the audience through the data narrative.